PO Box 65 070 Mairangi Bay, Auckland 0754, New Zealand
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www.keyindustries.co.nz

Safety Data Sheet: Strike Out

1. IDENTIFICATION

Product Name:	Strike Out
Other Names:	Environmentally Hazardous Substance, Liquid, NOS (contains cypermethrin)
Recommended Use:	Insecticide
Distributor:	Key Industries Ltd
Address:	PO Box 65 070 Mairangi Bay, Auckland 0754
Telephone:	09 917 1791
Emergency Phone:	0800 CHEMCALL (0800 243 622)
National Poisons Centre:	0800 764 766

2. HAZARDS IDENTIFICATION

Signal Word: DANGER

Hazards:	**Hazard Phrases**
Acute toxicity: oral category 4	Harmful if swallowed.
Eye damage category 2	Causes serious eye damage.
Skin sensitization category 1	May cause an allergic skin reaction.
Reproductive toxicity category 2	Suspected of damaging fertility or the unborn child
Specific target organ toxicity (repeated exposure) Category 1	Causes damage to organs through prolonged or repeated exposure.
Aquatic toxicity (acute/chronic) category 1	Very toxic to aquatic life with long lasting effects.
Hazardous to terrestrial vertebrates	
Hazardous to terrestrial invertebrates	

Precautionary Phrases:
If medical advice is needed, have product container or label at hand.
Keep out of reach of children.
Read label before use.
Obtain special instructions before use.
Do not handle until all safety precautions have been read and understood.
Do not breathe mist/spray
Wash any areas of the body that have come into contact thoroughly after handling.
Do not eat, drink or smoke when using this product.
Contaminated work clothing should not be allowed out of the workplace.
Wear protective gloves/eye protection/face protection.
Use personal protective equipment as required.
IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
IF ON SKIN: Wash with plenty of soap and water.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
IF exposed or concerned: Get medical advice/ attention.
Get medical advice/attention if you feel unwell.
Rinse mouth.
If skin irritation or rash occurs: Get medical advice/attention.
If eye irritation persists: Get medical advice/attention.
Take off contaminated clothing and wash before re-use.
Collect spillage.
Store locked up.
Dispose of contents/container in accordance with local/regional/national regulations
Take all reasonable steps to ensure that the substance does not cause any significant adverse effects to the environment beyond the application area
Do not apply directly into or onto water
Do not apply substance to plants if bees are foraging; or the plants are in flower or part flower

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Pictograms:

3. COMPOSITION: Information on Ingredients

Ingredient	CAS Number	Concentration (%w/w)
Cypermethrin	52315-07-8	19.5
Propylene glycol	57-55-6	<10
Dipropylene glycol methyl ether	34590-94-8	<5
Balance – ingredients of proprietary nature	Proprietary	To 100%

4. FIRST AID MEASURES

Consult the National Poisons Information Centre 0800 POISON (0800 764 766) or a doctor in every case of suspected chemical poisoning. Never give fluids or induce vomiting if a patient is unconscious or convulsing regardless of cause of injury. If breathing difficulties occur seek medical attention immediately.

Swallowed
If swallowed, do not induce vomiting. If conscious and alert, rinse mouth and drink 1-2 cupsful of water. Begin artificial respiration if the victim is not breathing. Use mouth to nose rather than mouth to mouth. Obtain medical attention.

Skin Contact
Immediately wash with plenty of soap and water. Remove contaminated clothing and shoes. Wash clothing before reuse. Destroy contaminated shoes and other contaminated leather articles such as belts and watchbands. Seek medical attention if irritation occurs.

Eye Contact
Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Remove contact lenses if present and easy to do and continue rinsing. Do NOT allow victim to rub eyes or keep eyes closed. Obtain urgent medical attention.

Inhalation
Move the victim to fresh air immediately. Begin artificial respiration if breathing has stopped. Obtain medical attention immediately.

First Aid facilities
Provide eye baths and safety showers close to areas where exposure may occur.

Medical Attention
The decision to induce vomiting should be made by a physician because rapid absorption of this material can occur through the lungs after aspiration from vomiting. Gastric lavage may be indicated if ingested. Emesis should not be performed unless a near fatal dose of material has been ingested. Emesis and aspiration of the lungs may cause chemical pneumonia with hydrocarbon solvent present. Treat symptomatically and supportively, monitoring the development of hypersensitivity reactions with respiratory distress. No known antidote. Do not confuse with cholinesterase poisoning. Skin contacted may be carefully cleaned with cleansing milk. Symptoms can be partially alleviated by the application of a vitamin E or moisturising cream or anaesthetic ointment. For eyes, instil local anaesthetic drops e.g. 1% amethocaine hydrochloride eye drops. Give analgesics as necessary. In all cases consult the National Poisons Centre for the most up to date treatment information.

5. FIRE FIGHTING MEASURES

Shut off product that may ‘fuel’ a fire if safe to do so. Allow trained personnel to attend a fire in progress, providing fire fighters with this Safety Data Sheet. Prevent extinguishing media from escaping to drains and waterways.

Suitable extinguishing media
Water fog, fine spray, alcohol resistant foam, dry chemical powder or carbon dioxide. Do not use water stream directly as this may spread fire or give a violent steam eruption. Cool fire exposed container with water spray.

Hazards from combustion products
Decomposition from combustion will emit acrid smoke and toxic fumes containing carbon oxides, nitrogen oxides, hydrogen chloride and hydrogen cyanide.

Precautions for fire fighters and special protective equipment
Full protective clothing with chemical goggles, butyl or neoprene gloves and self-contained breathing apparatus

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Hazchem Code
2Z

Flash Point
>95°C

6. ACCIDENTAL RELEASE MEASURES

Emergency Procedures

Prevent product from escaping to drains and waterways. Contain leaking packaging in a containment vessel or bunded area. Prevent any vapours or dust from building up in confined areas. Ensure that drain valves are closed at all times. Clean up minor spills immediately.

Methods and materials for containment for a major spill

Eliminate sources of ignition. Warn occupants of downwind areas of possible hazards. Keep the public away from the area. Prevent product from entering sewers, watercourses, or low-lying areas. Shut off the source of the spill if safe to do so. Advise relevant authorities if substance has entered a watercourse or sewer or has contaminated soil or vegetation. Take measures to minimise the effect on the groundwater. If possible recover product using a pump paying attention to flammability hazards or absorbent material. Collect and seal in properly labelled containers for disposal. Consult an expert on disposal of recovered material and ensure conformity to local disposal regulations. In all instances due consideration must be given for First Aid Measures (Section 4), PPE requirements (Section 8), Stability and Reactivity (Section 10) for this material.

7. HANDLING AND STORAGE

Precautions for safe handling
Keep out of reach of children. This product is combustible. Avoid sources of ignition including smoking. Keep containers closed. Use only in well-ventillated areas. When handling do not eat, drink or smoke. Wash hands thoroughly after handling. Wash clothing separately before reuse.

Conditions for safe storage
Store in a cool, dry place away from direct sunlight. Keep away from heat and ignition sources. Store in original containers. Protect from physical damage to prevent accidental release. Do not store with food, feedstuffs, fertilizers and seeds.

Incompatible materials
Natural Rubbers.

Fire Extinguisher Requirements
No specific requirements.

8. EXPOSURE CONTROLS: Personal Protection

Exposure Limits
A WES has been set by MBIE for two materials in this product, propane-1,2 diol (propylene glycol) and Propanol, 1(or 2)-(2-methoxymethylethoxy) (dipropylene glycol methyl ether). The WES for propylene glycol is TWA (vapour and particulates) 150ppm or 474 mg/m^3. TWA (particulates only) 10 mg/m^3. The WES for dipropylene glycol methyl ether is TWA 100ppm or 606 mg/m^3, STEL 150ppm or 909 mg/m^3

Engineering Controls:
The use of local exhaust ventilation is recommended to control process emissions near the source. Sufficient ventilation should be provided to keep the solvent in air concentrations below any relevant exposure limit. Provide mechanical ventilation of confined spaces. Explosion proof electrical equipment recommended but not required based on the flashpoint of this material.

Hygiene Controls:
Facilities storing or utilising this material should be equipped with an eyewash facility, safety shower and facility for washing hands/face after work.

Personal Protective Equipment
Respiratory Protection: An approved half/full face filter respirator for organic vapours should be worn when applying this product.
Eye protection: Always use safety glasses or a chemical goggles when handling this product. Contact lenses may absorb and concentrate irritants, glasses are recommended.

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Skin/ Body Protection: Always wear long sleeves and long trousers or coveralls, and enclosed footwear of safety boots when handling this product. It is recommended that chemical resistant gloves (eg nitrile, neoprene) be worn when handling this product.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	Semi-opaque pale yellow to brown liquid
Odour:	Characteristic odour
Odour Threshold:	No data available
Boiling Point (°C):	Approximately 100
Boiling Point Range	No data available
Melting Point (°C):	No data available
Flash Point (°C):	>95
Lower Explosive Limit, LEL (%):	No data available
Upper Explosive Limit, UEL (%):	No data available
SG/ Density, 20°C (g/mL):	1.025 – 1.055
Vapour Pressure, 20°C (kPa):	No data available
Vapour Density:	No data available
Alkalinity/ acidity as pH:	4.5 – 6.0
Solubility in water:	Forms a emulsion
Partition Coefficient (Kow):	No data available for formulated product
Auto-ignition Temperature (°C):	No data available
Decomposition Temperature (°C):	No data available
Kinematic Viscosity:	No data available

The values listed are indicative of this product's physical and chemical properties. No data is available for any properties not listed above.

10. STABILITY AND REACTIVITY

Chemical stability
Stable at room temperature and pressure. Separates at elevated temperatures but recombines at ambient temeprature
Hazardous decomposition products
Decomposition from combustion will emit acrid smoke and toxic fumes containing carbon oxides, nitrogen oxides, hydrogen chloride and hydrogen cyanide.
Specific Materials to Avoid
Strong acids, alkalis, oxidising agents, reducing agents and heat.
Hazardous Polymerisation
Not known to occur.

11. TOXICOLOGICAL INFORMATION

Acute Effects
Ingestion: Moderate toxicity if swallowed. Symptoms include burning sensations in the mouth, headache, dizziness, drowsiness, diarrhoea, nausea, vomiting, listlessness, stomach pain, muscular twitching or arms or legs.
Ingestion of small amounts is unlikey to cause permanent injury, however swallowing large quantities can result in confusion, convulsions, coma, seizures and death. Aspiration of the lungs can occur during vomiting leading to lung damage or death due to chemical pneumonitis. Contact a doctor immediately.
Eye Contact: The liquid is irritating to the eyes and may result in corneal injury. Avoid eye contact.
Skin Contact: Irritating. Contact with the skin, especially the face, may result in initial stinging or burning sensations (parasthesia) followed by numbing which may persist for a few hours. Avoid skin contact.
Inhalation: The vapour/mist is discomforting to the upper respiratory tract and lungs. Acute effects from inhalation of high vapour concentrations may cause effects similar to that of ingestion.
Chronic Effects
Prolonged or repeated exposure may result in central nervous system effects such as tremors, uncoordination and disorientation. Prolonged or repeated skin contact may result in the development of sensitisation (allergy). The

active ingredient, cypermethrin is suspected of damaging fertility from repeated oral exposure. Persons with pre-existing conditions are advised to limit or avoid product contact.

Other Health Effects Information

Not Available.

Toxicological Information

Oral LD_{50}: 138 mg/kg (mouse), cypermethrin active ingredient; 642 mg/kg (product, estimated)
Dermal LD_{50}: 1600 mg/kg (rat), cypermethrin active ingredient; >5000 mg/kg (product, estimated)
Inhalation LC_{50}: 2.5 mg/L (rat), cypermethrin active ingredient; >10 mg/L (product, estimated)

12. ECOLOGICAL INFORMATION

Ecotoxicity

This product is highly toxic to the environment, with the potential to result in irreparable and irreversible effects. See hazard classifications in section 15 of this document. No specific ecological data available on this product.
Persistence/ Biodegradability: log P: 5.3 (cypermethrin). This substance is regarded as non bioaccumulative and non-persistant by EPA.
Mobility: This product is not readily soluble with water limiting its mobility in the environment. This product is likely to have low mobility in the environment and low leaching potential

Environmental Toxicity:

Fish toxicity LC_{50}: 0.0004 mg/L; cypermethrin active ingredient
Daphnia Magna EC_{50}: 0.00036 mg/L; cypermethrin active ingredient
Honeybee: 0.02 µg/bee; cypermethrin active ingredient

13. DISPOSAL CONSIDERATIONS

Product Disposal

Dispose of product only by using according to label or using an approved waste disposal contractor. If this material as supplied becomes a waste care should be taken to ensure compliance with national and local authorities. It is the responsibility of the waste generator to determine the toxicity and physical properties of the waste generated to determine the proper waste identification and disposal methods in compliance with applicable regulations. Do not dispose of via municipal sewers, drains, natural streams or rivers

Packaging Disposal

Triple rinse container and add rinsate to the spray tank. Empty packaging should be taken for recycling, recovery or disposal through a suitably qualified or licensed contractor. Care should be taken to ensure compliance with national and local authorities. Packaging may still contain product residue that may be harmful. Incinerate via approved incinerators or crush and bury in an approved landfill. Ensure that empty packaging is managed in accordance with Dangerous Goods and HSNO regulations.

14. TRANSPORT INFORMATION

UN No:	UN3082
Proper Shipping Name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, NOS (contains cypermethrin)
DG Class:	9
Packing Group:	III
Hazchem Code:	3Z

Dangerous Goods Segregation

This product is classified as Dangerous Goods by Road and Rail. Please consult the Land Transport Rule: Dangerous Goods 2005, and NZS 5433:1999 Transport of Dangerous Goods on Land for further information.

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15. REGULATORY INFORMATION

Country/ Region: New Zealand

ACVM Approval Number:	Not Applicable
EPA Approval Number:	HSR100346
HSNO Controls:	Trigger Quantities for this Material: • Certified Handler Test Certificate: Not required • Location/Transit Depot Certificate: Not Required • Hazardous Atmousphere Zone: Not Required • Signage: 100 L • Emergency Plan, Secondary Containment: 100L • Tracking: Not required • Record Keeping: Required for application of ≥3kg of product in 24 hours (See HPC Notice 2017, section 48 for more detail). The trigger quantities above must take into account any other hazardous substance that is present at that location. This represents a partial list of the controls for this material. Contact Key Industries Ltd a full list of EPA controls.

16. OTHER INFORMATION

Reasons for Issue:
Updated for GHS compliance.
Abbreviations:
TWA - the highest allowable exposure concentration in an eight-hour day for a five-day working week
STEL - maximum allowable exposure concentration at any time
EPA - Environmental Protection Agency of New Zealand
HSNO - Hazardous Substances and New Organisms
MBIE - Ministry of Business, Innovation and Employment
References:

- Supplier Safety Data Sheets
- Hazardous Substances Databank
- EPA Chemical Classification Information Database
- FOOTPRINT Pesticides Database
- Synapse Chemlib Handbook of Solvents